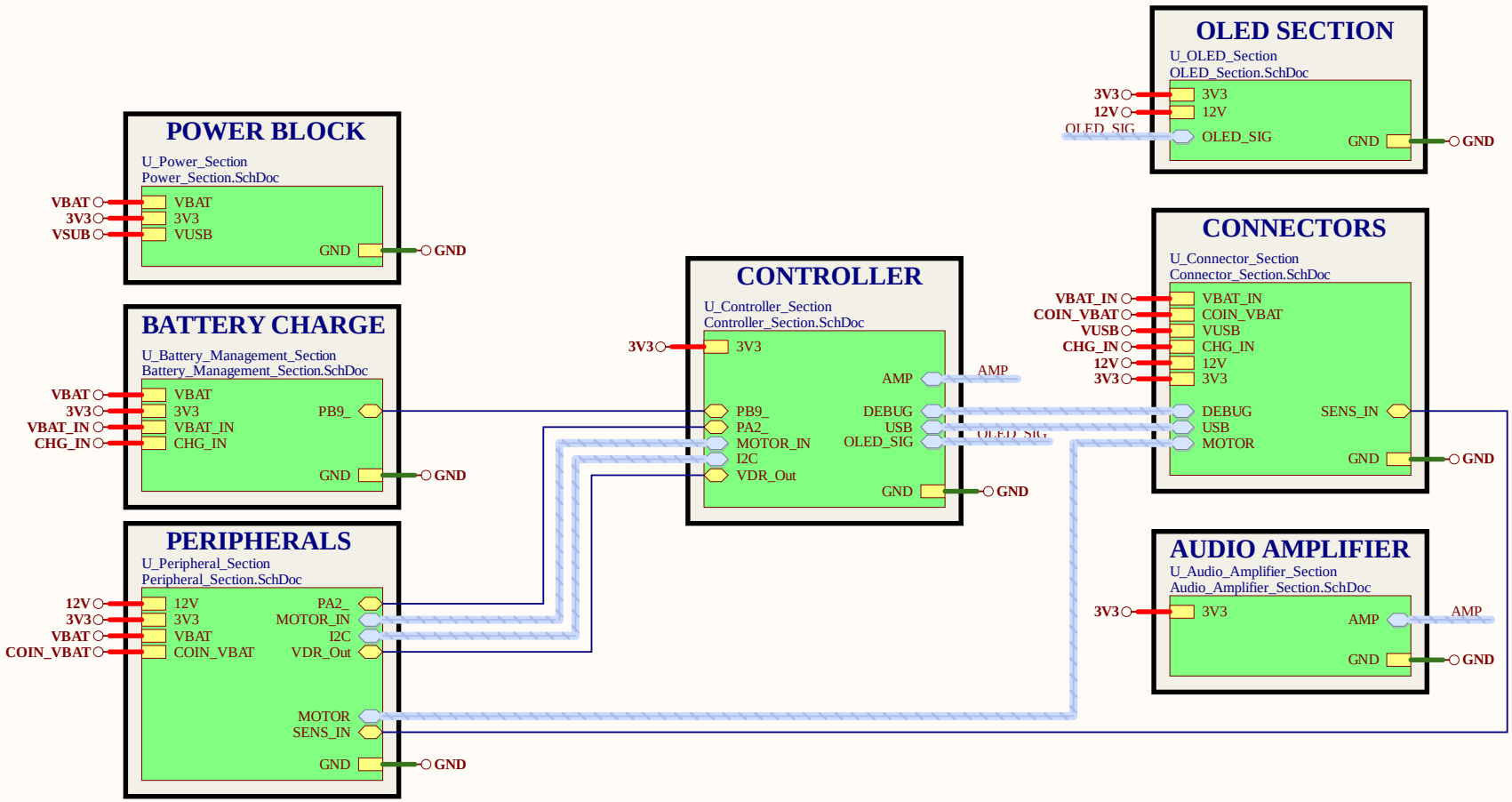
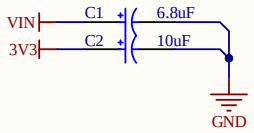
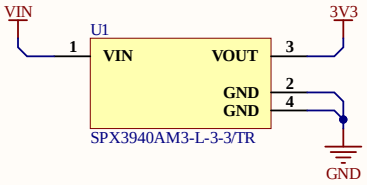


Title	Grantiot.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	3/09/2022	Sheet 1 of 7	
File:	E:\Git\..\Top Sheet.SchDoc	Drawn By:	Huzaifa Ali

DECOUPLING CAPS



MCU VOLTAGE REGULATOR IC



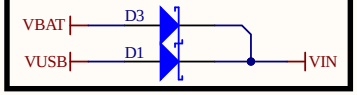
TEST POINT



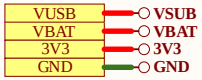
POWER INDICATION LED



ORING CIRCUIT



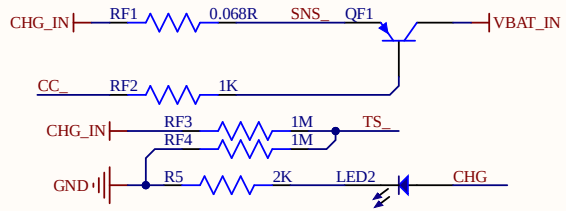
VBAT=6V
VUSB=5V
VF of Diode= 360mV @1A
VR=30V
IF=1A



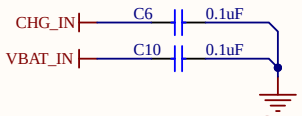
Title	Grantiot.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	3/09/2022	Sheet2 of	7
File:	E:\Git\..\Power_Section.SchDoc	Drawn By:	Huzaifa Ali

PB9_ PB9_

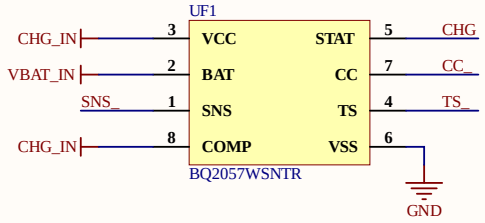
BATTERY CHARGING IC CIRCUIT



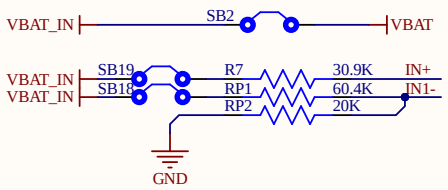
DECOUPLING CAPS



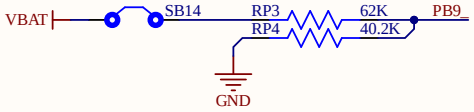
BATTERY CHARGING IC



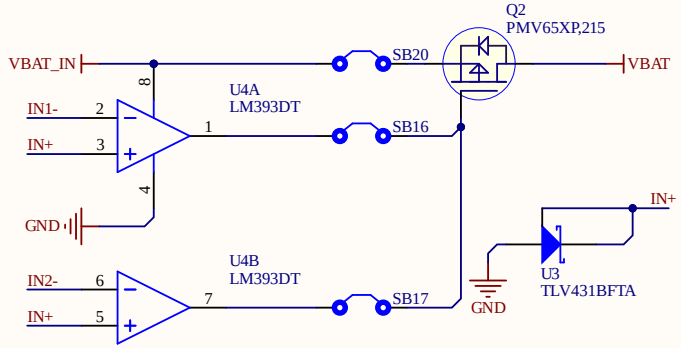
Divider for Battery Cutoff at 5V (AA Cell)



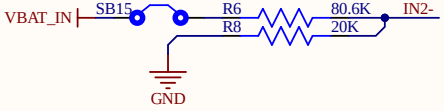
Divider for Battery Voltage Monitoring

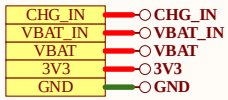


COMPARATOR IC CIRCUIT

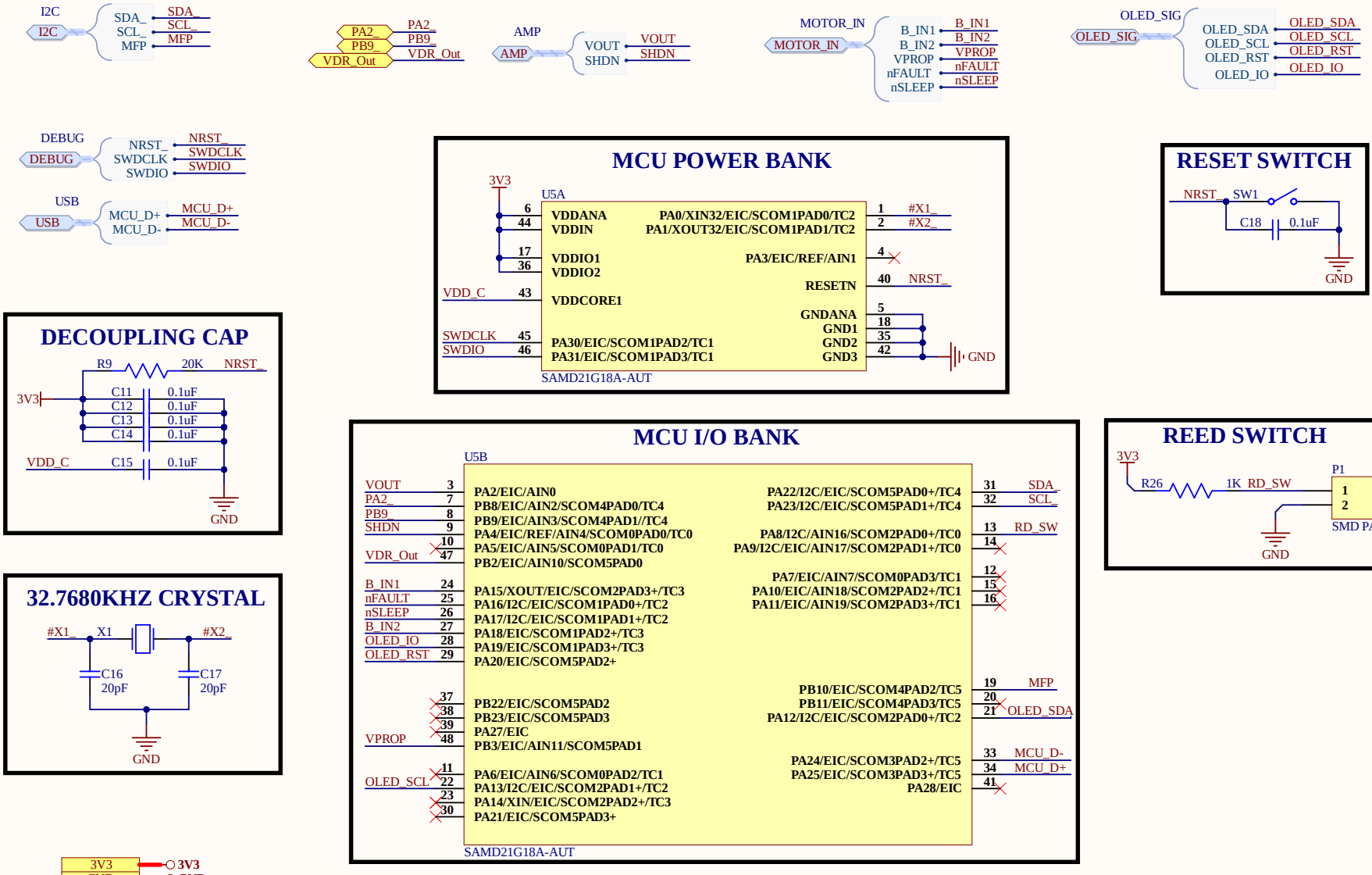


Divider for Battery Cutoff at 6.2V (LiPo Battery)



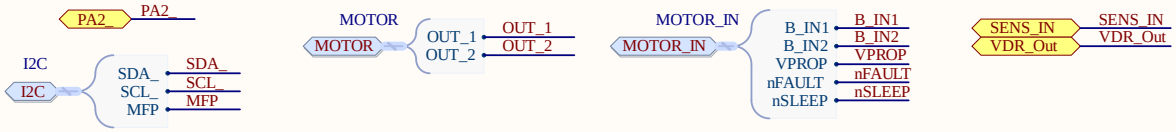


eSpark Consulting Group	Title: Grantiot.PrjPcb	
	Size: A4	Number: 1
	Date: 3/09/2022	Revision: 1
	File: E:\Git\..\Battery_Management_Section.SchDoc	Sheet 3 of 7
		Drawn By: Huzaifa Ali

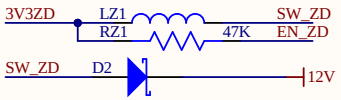


Controller Section

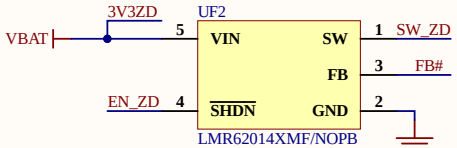
Title	Grantiot.PrjPcb		
Size	A4	Number	1
Revision	1		
Date:	3/09/2022	Sheet 4 of	7
File:	E:\Git\..\Controller_Section.SchDoc	Drawn By:	Huzaifa Ali



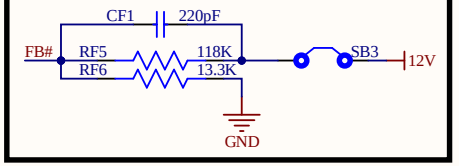
SUPPORTING CIRCUIT



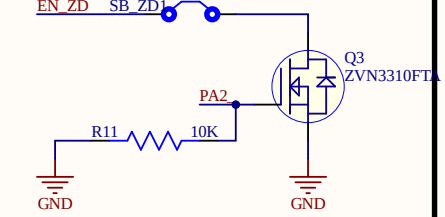
12V BOOST REGULATOR



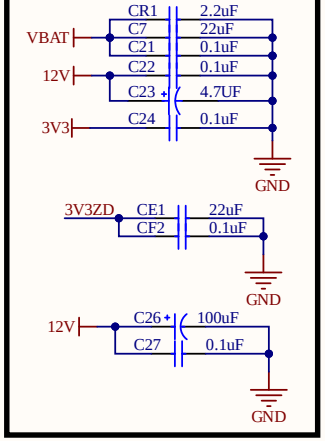
FEEDBACK NETWORK



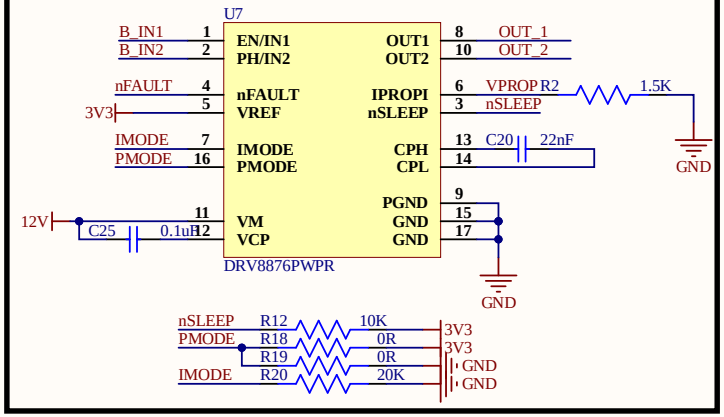
FEEDBACK NETWORK



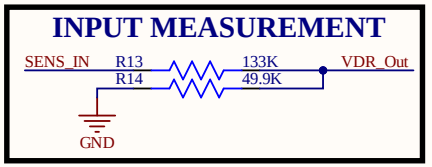
DECOUPLING CAPS



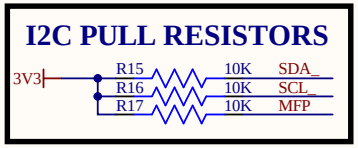
MOTOR DRIVER IC



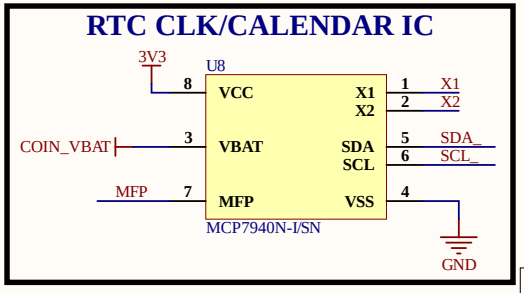
INPUT MEASUREMENT



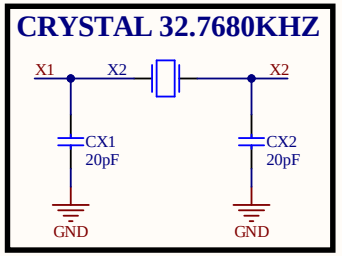
I2C PULL RESISTORS

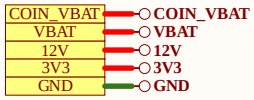


RTC CLK/CALENDAR IC



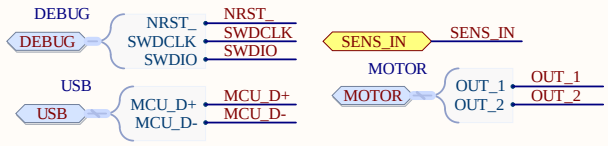
CRYSTAL 32.7680KHZ



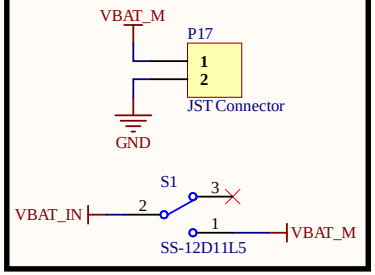


Peripheral Section

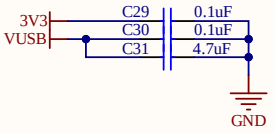
eSpark Consulting Group	Title: Grantiot.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 3/09/2022	Sheet 6 of 7	
	File: E:\Git\..\Peripheral_Section.SchDoc	Drawn By: Huzaifa Ali	



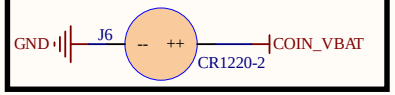
BATTERY INPUT



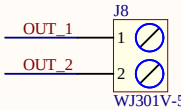
DECOUPLING CAP



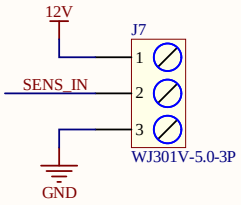
COIN CELL CONNECTOR



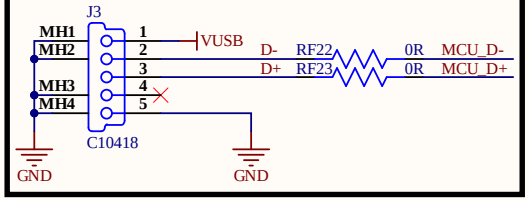
MOTOR OUTPUT



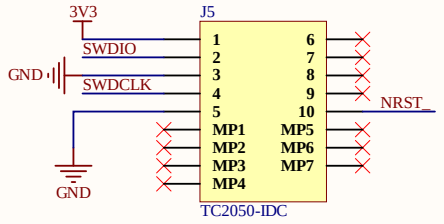
12V MOTOR SUPPLY



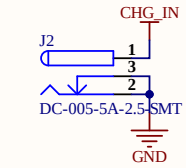
USB DATA TRANSFER



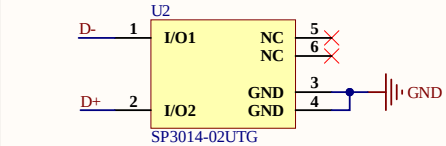
JTAG DEBUGGING CONNECTOR



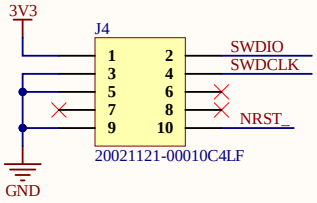
CHARGING ADAPTER



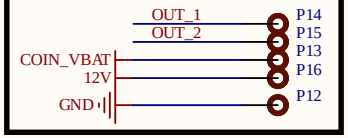
TVS ARRAY FOR USB INPUT

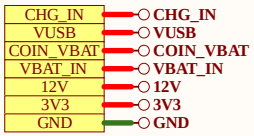


SERIAL WIRE DEBUGGING HEADER



TEST POINTS





Connector Section

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Size: A4	Number: 1		Revision: 1
Date: 3/09/2022		Sheet 7 of 7	
File: E:\Git\..\Connector_Section.SchDoc		Drawn By: Huzaifa Ali	

AMP

AMP

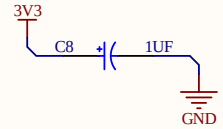
VOUT

SHDN

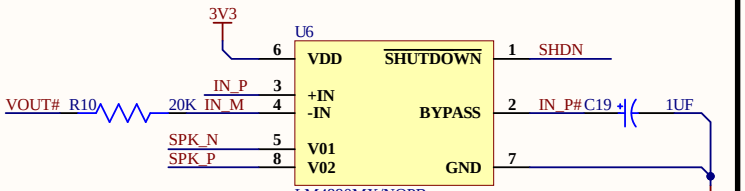
VOUT

SHDN

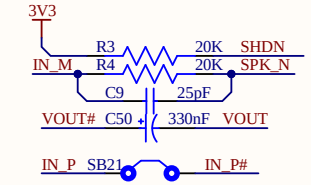
DECOUPLING CAP



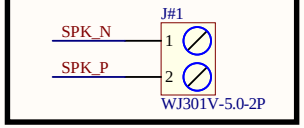
1W AUDIO AMPLIFIER IC



SUPPORTING CIRCUIT

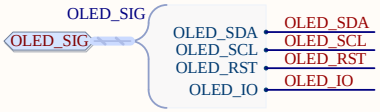


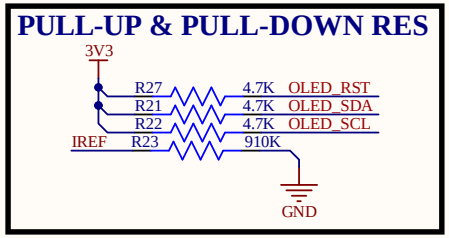
SPEAKER OUTPUT

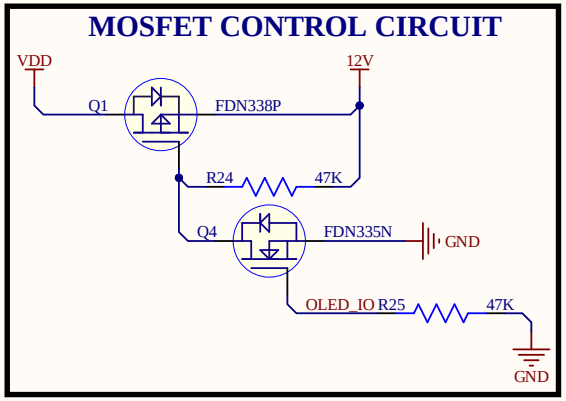


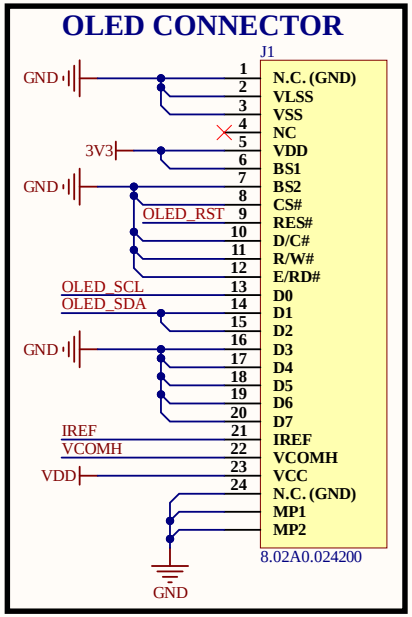


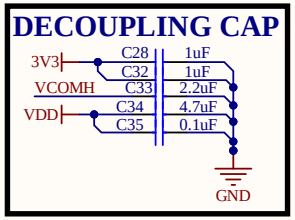
Title	Grantiot.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	3/09/2022	Sheet* of	*
File:	E:\Git\..\Audio_Amplifier_Section.SchDoc	Drawn By:	Huzaifa Ali





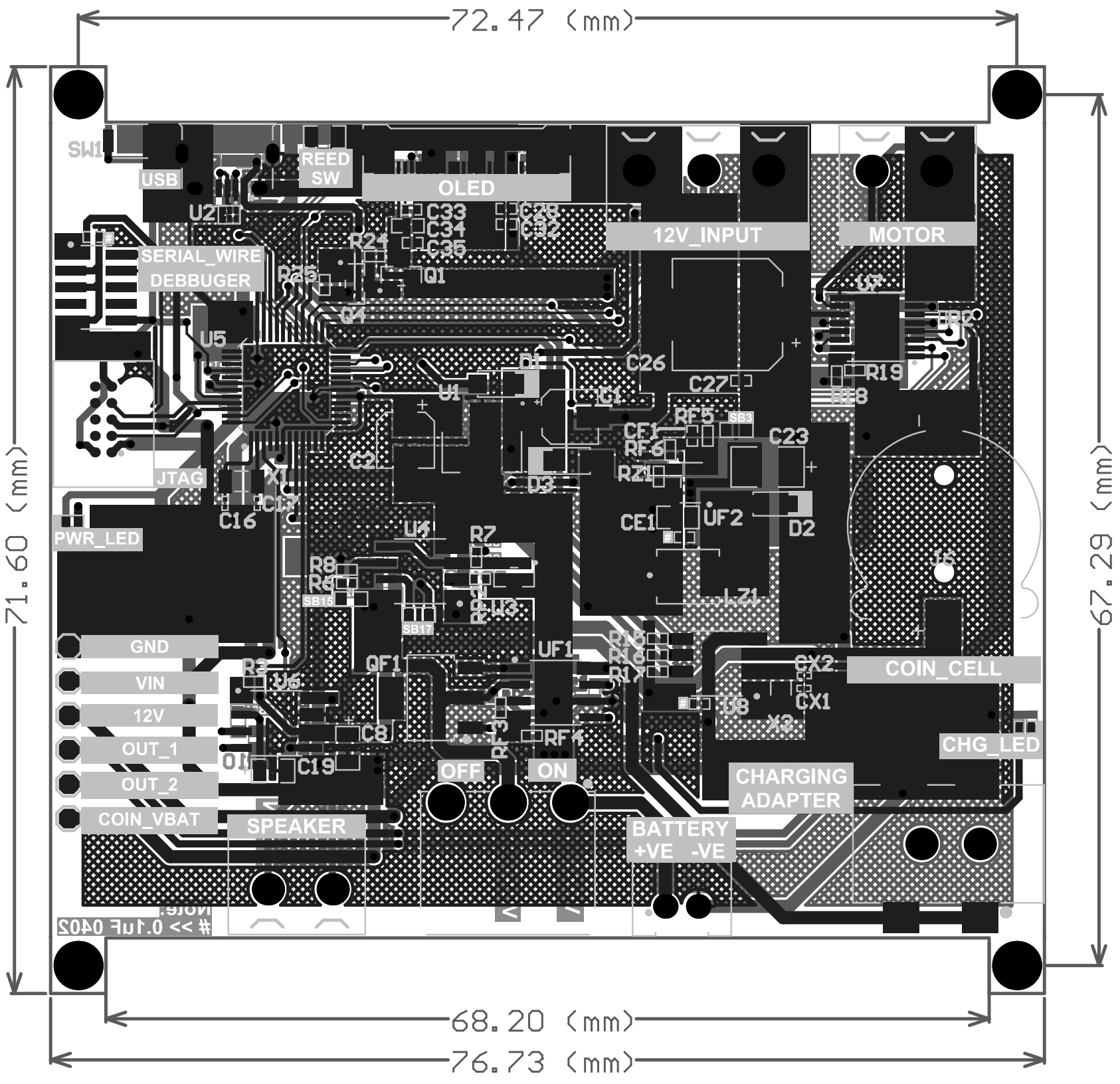








Title: Grantiot.PrjPcb			
Size: A4	Number: 1		Revision: 1
Date: 3/09/2022		Sheet* of *	
File: E:\Git\..\OLED_Section.SchDoc		Drawn By: Huzaifa Ali	


72.47 (mm)
71.60 (mm)
67.29 (mm)
68.20 (mm)
76.73 (mm)
SW1
USB
REED SW
OLED
U2
C33
C34
C35
C28
C32
12V_INPUT
MOTOR
R24
R25
Q1
Q4
SERIAL_WIRE DEBBUGER
U5
U1
D1
C26
C27
C1
R19
U8
RF5
SB3
CE1
RF6
C23
C2
RZ1
D3
JTAG
X1
C16
C17
CE1
UF2
D2
PWR_LED
U4
RZ
R8
R6
SB15
U3
SB17
J6
Z1
GND
VIN
12V
OUT_1
OUT_2
COIN_VBAT
UF1
QF1
R3
U6
R15
R16
R17
CX2
CX1
U8
X2
COIN_CELL
C8
C19
RF4
CHG_LED
OFF
ON
CHARGING ADAPTER
SPEAKER
BATTERY +VE -VE